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Iterations Statements

↳ loops

loops are used to repeat a set of instructions multiple times until a condition is satisfied.

Types of loops

↳ 1) While loop

↳ 2) Do-while loop

↳ 3) For-loop

1) while loop

Runs while condn is true

syntax:

```
while (expression)
{
    // do something;
}
```

eg: int i = 1;
while (i<=5) { (boolean)
    s.o.p (i);
    i++;
}

2) Do-while loop
Executes at least once
syntax:
do {
    // do something
} while (expression)

eg:
do {
    s.o.p (i);
    i++;
} while (i<=10);

3. For loop
used when number of iterations are known
syntax:
for (initialization; condition; inc/decr)
{
}

eg: for (int i=1; i<=10; i++) (boolean → true/false)
{
    s.o.p(in (i));
}

Flow of control for for
1) first assignment statement is executed (var def)
2) then second conditional stmt is evaluated (true/false)

3) If true, ~~cell~~ control flow will evaluate the body of the loop
4) once loop body is finished, control flow will go back to the for statement and third ~~statement~~ increment statement will be evaluated.
5) Again, conditional statement is evaluated
6) Repeat 2 - 5.

Comma separated variation

```
cin i=0 ; j=1 ; i<=n ; i++)
```

Nested loops

```
for (int i=1 ; i<=5 ; i++)
{
    for (int j=1 ; j<=5 ; j++) {
        // do something
    }
}
```

Patterns

```
*
* *
* * *
* * * *
* * * * *
```

outer loop → for no. of lines
no. of stars → inner loop

```
for (int i=1 ; i<=5 ; i++)
{
    for (int j=1 ; j<=i ; j++)
    {
        s.o.p("*")
    }
    s.o.p(n);
```

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* Jump statements in Java

break continue

break → exit from loop dynamically.

eg: whether a number is prime or not

```
int p = 9;
// 2, 3, 4, 5 ... 9
for (int i = 2; i < p; i++)
    if (p % i == 0)
        s.o.p ("number is not prime");
    }   break;
}
if (i == p) {
    s.o.p ("number is prime");
}
```

continue → means go to next iteration

* Code blocks

```
First : {
    second : {
        third : {
            s.o.p ("Hello");
        }   break first;
```

Patterns Programs solved